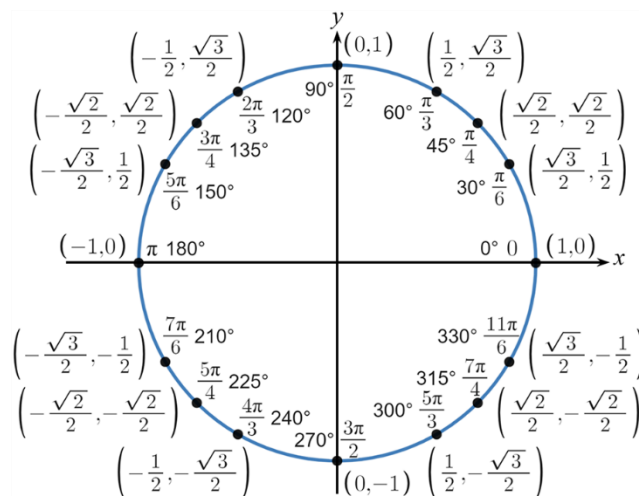


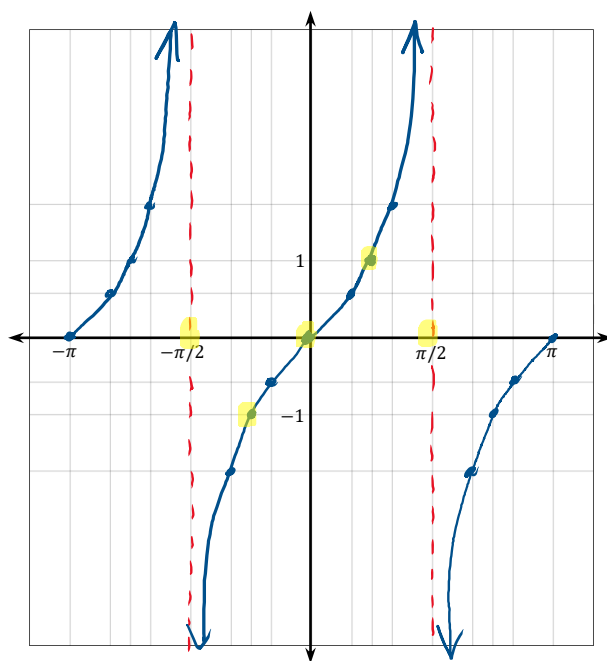
Graphing Other Trigonometric Functions

Graphing Tangent and Cotangent

We will again refer to the unit circle to construct tangent and cotangent. As before, we consider the angle θ as the input and graph this independent variable along the horizontal axis. The corresponding output, y/x -values for tangent and x/y -values for cotangent from the unit circle, will be graphed along the vertical axis as usual. We'll place a vertical asymptote in locations where the function values are undefined.



$$y = \tan x$$



Domain: $x \neq \frac{\pi}{2} + \pi k, k \text{ any integer}$

Range: $(-\infty, \infty)$

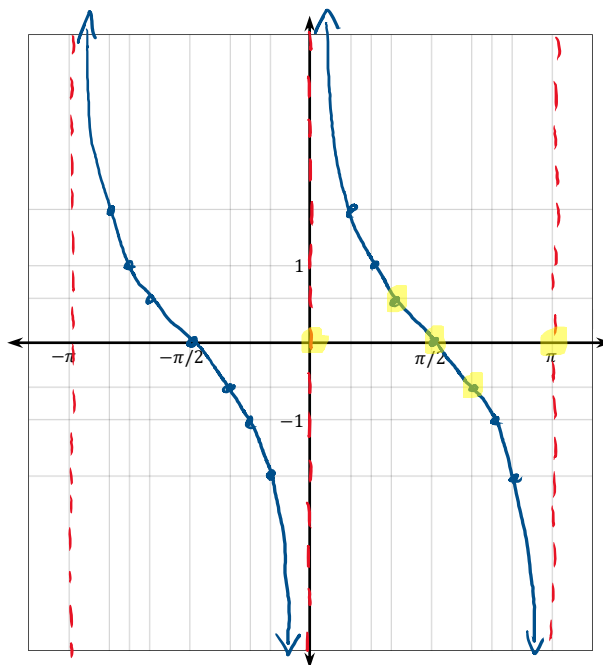
Amplitude: n/a

Period: π

y-intercepts: $(0, 0)$

x-intercepts: $(\pi k, 0), k \text{ any integer}$

$$y = \cot x$$



Domain: $x \neq \pi k, k \text{ any integer}$

Range: $(-\infty, \infty)$

Amplitude: n/a

Period: π

y-intercepts: none

x-intercepts: $(\frac{\pi}{2} + \pi k, 0), k \text{ any integer}$

Combining Transformations

IMPORTANT INFO: Summary of Transformations

Consider $y = a \tan[b(x + c)] + d$ or $y = a \cot[b(x + c)] + d$, with $b > 0$.

Vertical Stretch/Compress:	The value $ a $ is the vertical stretch/compression. If $a < 0$, the graph will be reflected across the horizontal axis.
Period:	The value π/b is the period.
Horizontal Shift:	If $c > 0$, the graph is shifted left c units. If $c < 0$, the graph is shifted right $ c $ units. (Note some texts refer to this as phase shift.)
Vertical Shift:	If $d > 0$, the graph is shifted up d units. If $d < 0$, the graph is shifted down $ d $ units.

EXAMPLE 5

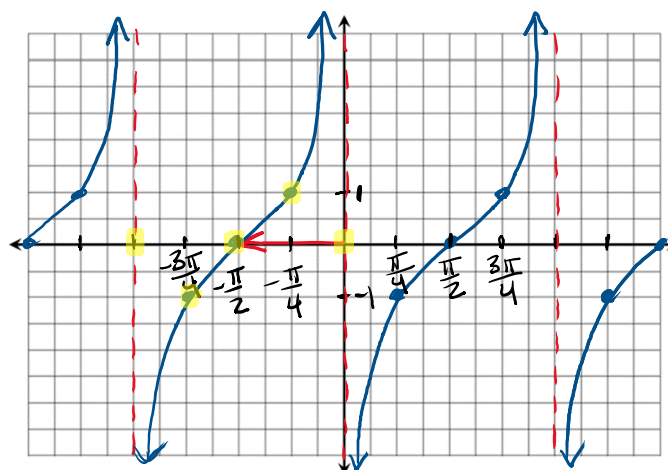
Graph each function below.

(a) $y = \tan\left(x + \frac{\pi}{2}\right)$

period = π

step = $\frac{\pi}{4}$

H. shift left $\frac{\pi}{2} = \frac{2\pi}{4}$

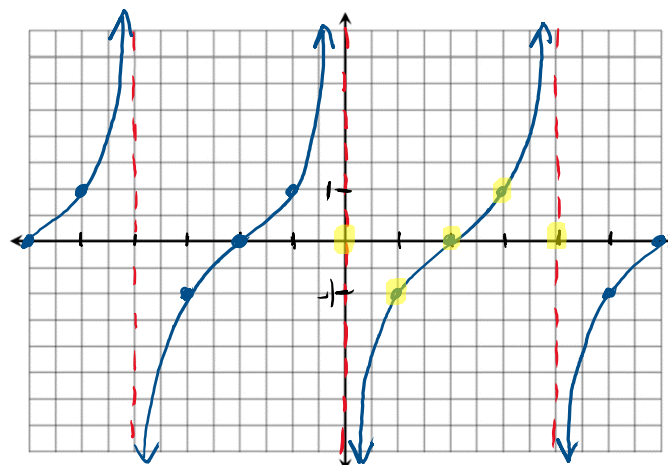


(b) $y = -\cot(x)$

period = π

step = $\frac{\pi}{4}$

reflect across
H. axis



SAME
GRAPH!

EXAMPLE 2

Match each graph with its corresponding equation.

$y = \tan(x)$ **D**

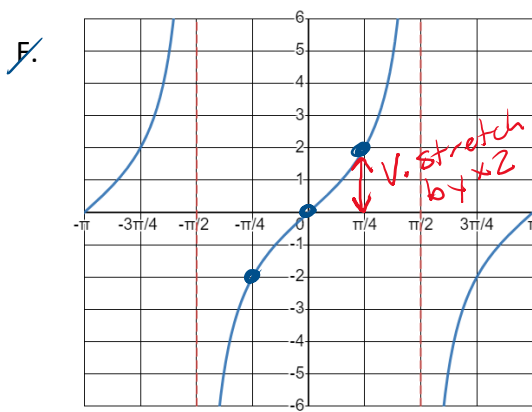
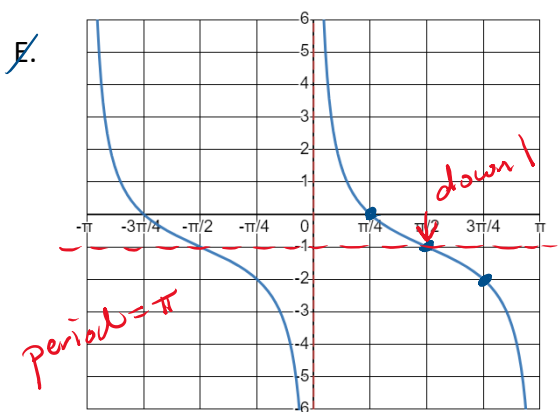
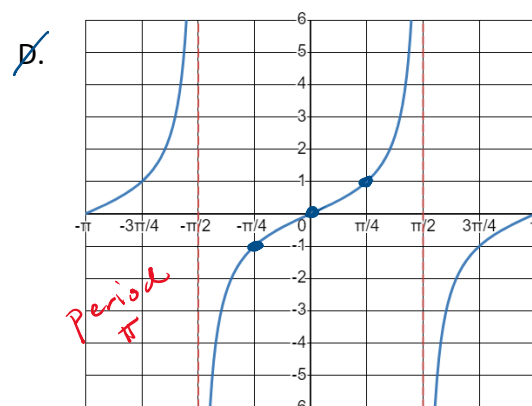
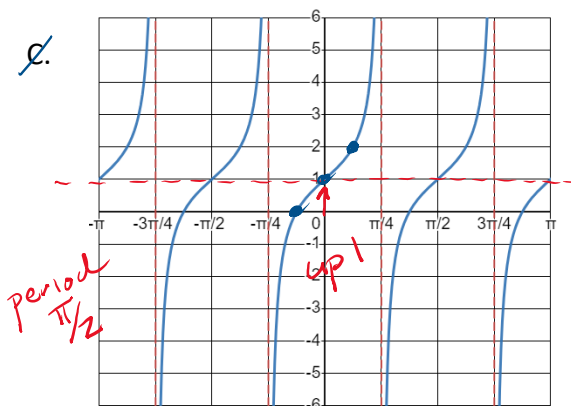
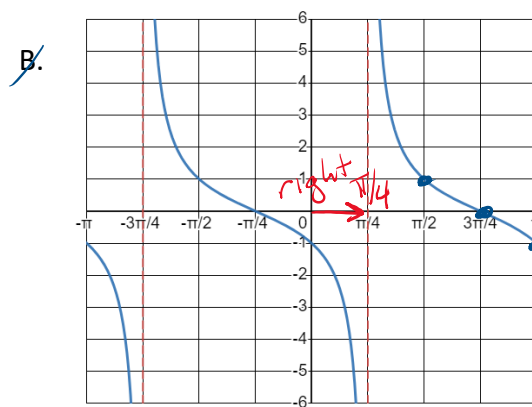
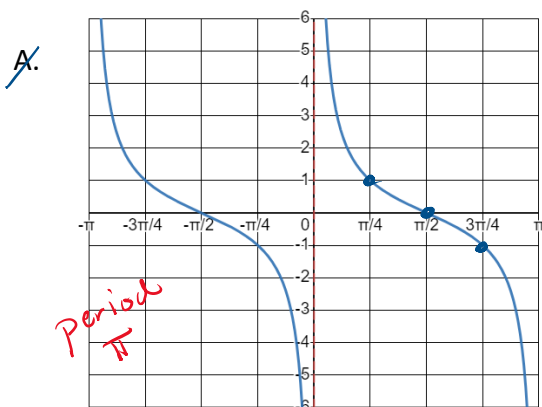
$y = 2 \tan(x)$ **F**

$y = \tan(2x) + 1$ **C**

$y = \cot(x)$ **A**

$y = \cot(x) - 1$ **E**

$y = \cot\left(x - \frac{\pi}{4}\right)$ **B**



Graphing Secant and Cosecant

After some practice with graphing sine and cosine, you'll be happy to know the graphs for cosecant and secant are closely related. These are reciprocal functions, and we can apply our knowledge of sine and cosine to construct these graphs. In fact, when asked to produce a sketch, we'll always begin by considering the related sine or cosine function first. Complete each of the following statements, then apply your conclusions to produce a sketch for cosecant and secant. Draw a vertical asymptote anywhere a function is undefined.

If $\sin \theta = 1$, then $\csc \theta = \underline{1}$.

If $\sin \theta = -1$, then $\csc \theta = \underline{-1}$.

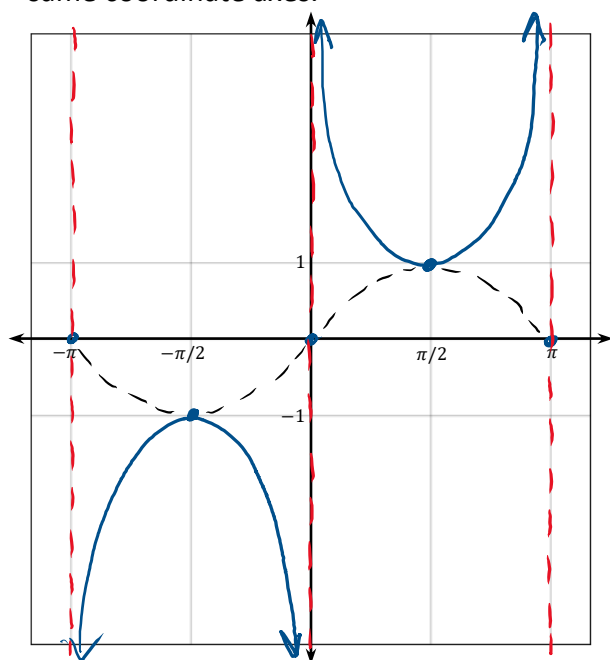
If $\sin \theta = 0$, then $\csc \theta = \underline{\text{undef.}}$

If $\cos \theta = 1$, then $\sec \theta = \underline{1}$.

If $\cos \theta = -1$, then $\sec \theta = \underline{-1}$.

If $\cos \theta = 0$, then $\sec \theta = \underline{\text{undef.}}$

Graph $y = \sin x$ and $y = \csc x$ on the same coordinate axes.



Domain: $x \neq \pi k, k \text{ any integer}$

Range: $(-\infty, -1] \cup [1, \infty)$

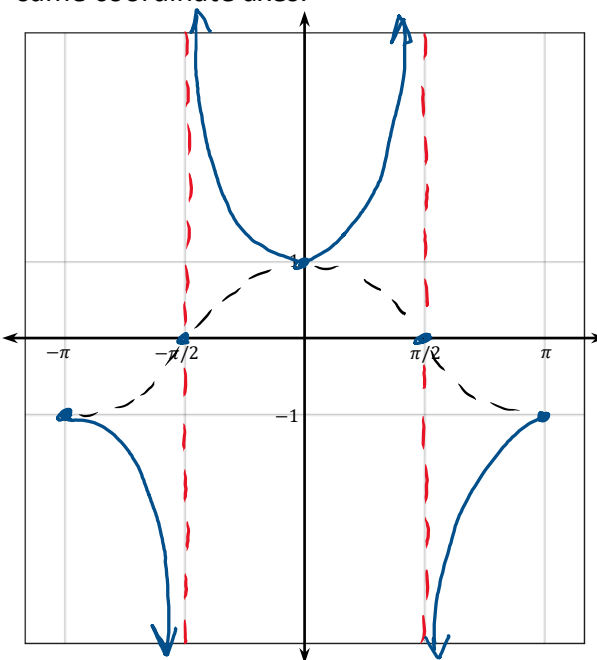
Amplitude: $\underline{n/a}$

Period: $\underline{2\pi}$

y-intercepts: $\underline{\text{none}}$

x-intercepts: $\underline{\text{none}}$

Graph $y = \cos x$ and $y = \sec x$ on the same coordinate axes.



Domain: $x \neq \frac{\pi}{2} + \pi k, k \text{ any integer}$

Range: $(-\infty, -1] \cup [1, \infty)$

Amplitude: $\underline{n/a}$

Period: $\underline{2\pi}$

y-intercepts: $\underline{(0, 1)}$

x-intercepts: $\underline{\text{none}}$

EXAMPLE 3

Use your knowledge of sine and cosine graphs to sketch each function below.

(a) $y = 2 \sec\left(x + \frac{\pi}{2}\right)$

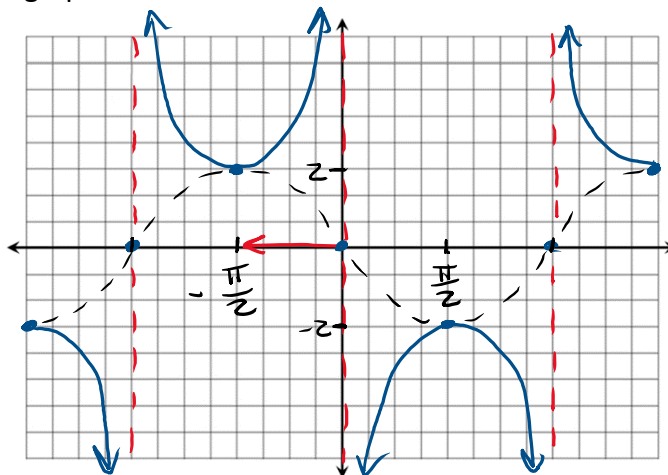
$y = 2 \cos\left(x + \frac{\pi}{2}\right)$

amp = 2

period = 2π

step = $\frac{2\pi}{4} = \frac{\pi}{2}$

H. shift left $\frac{\pi}{2}$



(b) $y = 1 - \csc(2x)$

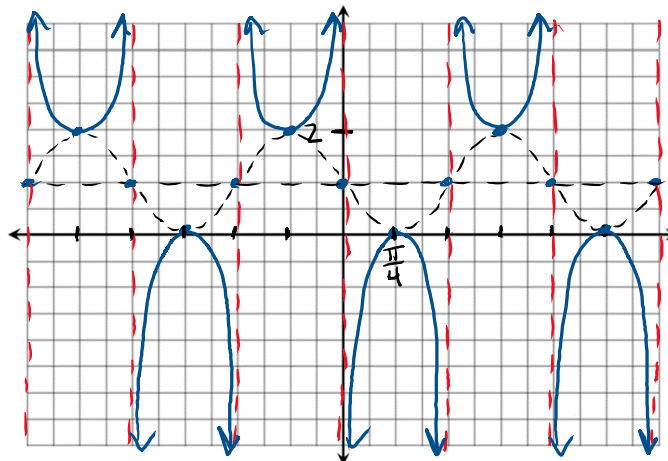
$y = 1 - \sin(2x)$

amp = 1, reflect across H. axis

period = $\frac{2\pi}{2} = \pi$

step = $\frac{\pi}{4}$

V. shift up 1



H. axis
 $y = 1$

EXAMPLE 4

Write an equation for the graph shown here.

amp = 1

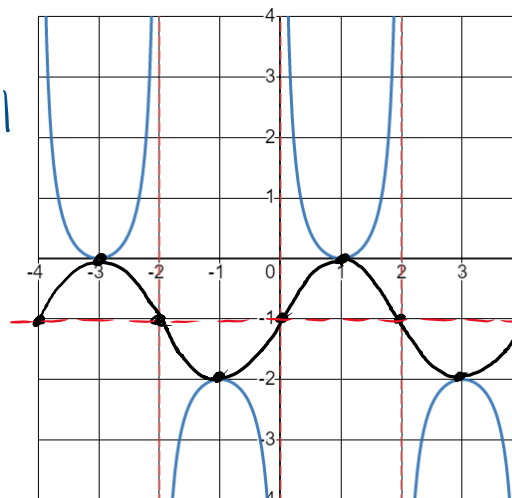
period = 4

V. shift down 1

$y = \sin\left(\frac{\pi}{2}x\right) - 1$

$y = \csc\left(\frac{\pi}{2}x\right) - 1$

$\frac{2\pi}{\pi/2} = 4 \checkmark$



H. axis
 $y = -1$

4

